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COMPAS : Unfair Algorithm ?

11 min read · Nov 22, 2023

Prathamesh Patalay Follow

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In this article, I try to visualise the findings from ProPublica's analysis[1] of the COMPAS recidivism algorithm, and responses to it that challenge ProPublica's claims. By visualising the findings, I hope to make it easier for readers to understand some of the nuances of why this research from ProPublica was widely discussed/debated and some of the reasons why the algorithm makes the kind of predictions it does.

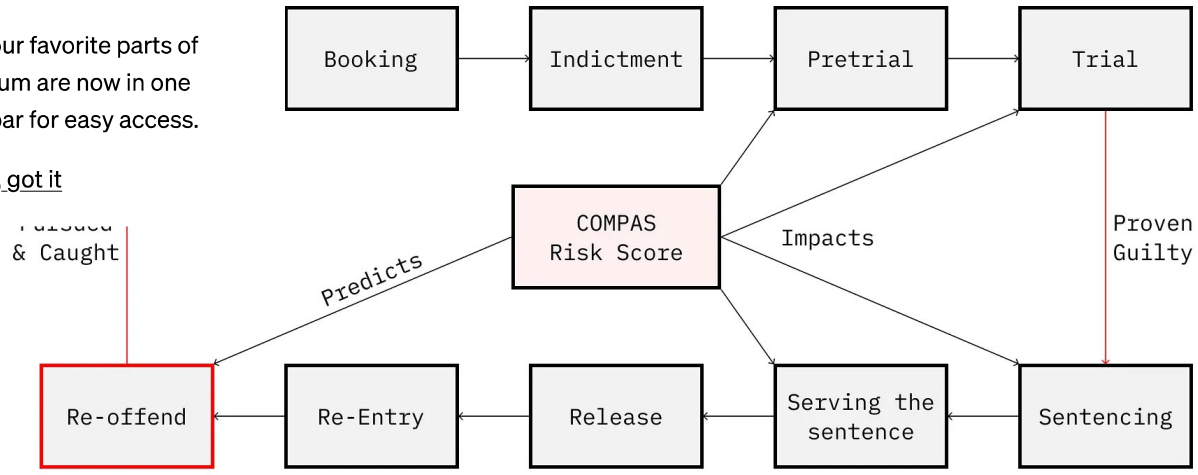
What is COMPAS

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) recidivism algorithm is a tool used in the US criminal justice system to assess the risk of an individual recidivating : reoffending after being previously arrested.

The tool generates a risk score for an individual, predicting the likelihood of them committing a crime in the future. This score is then used in the criminal justice process, to recommend probation conditions, corrective measures and type of treatment program. Although it is designed to support post-sentence supervision(not to decide criminal sentences),[3] many judges use it for criminal sentencing & it has been used in pretrial hearings too. [1] That's also true for Broward County, where ProPublica conducted the study.

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A simplified overview of steps in the criminal justice process[10] and how COMPAS scores relate to them

The ProPublica study

ProPublica, a non-profit investigative journalism newsroom, performed a study of the Risk scores assigned to 7000 people arrested in Broward County, Florida in 2013 and 2014[2]. From their findings they claimed that the COMPAS algorithm was biased against Black arrestees.

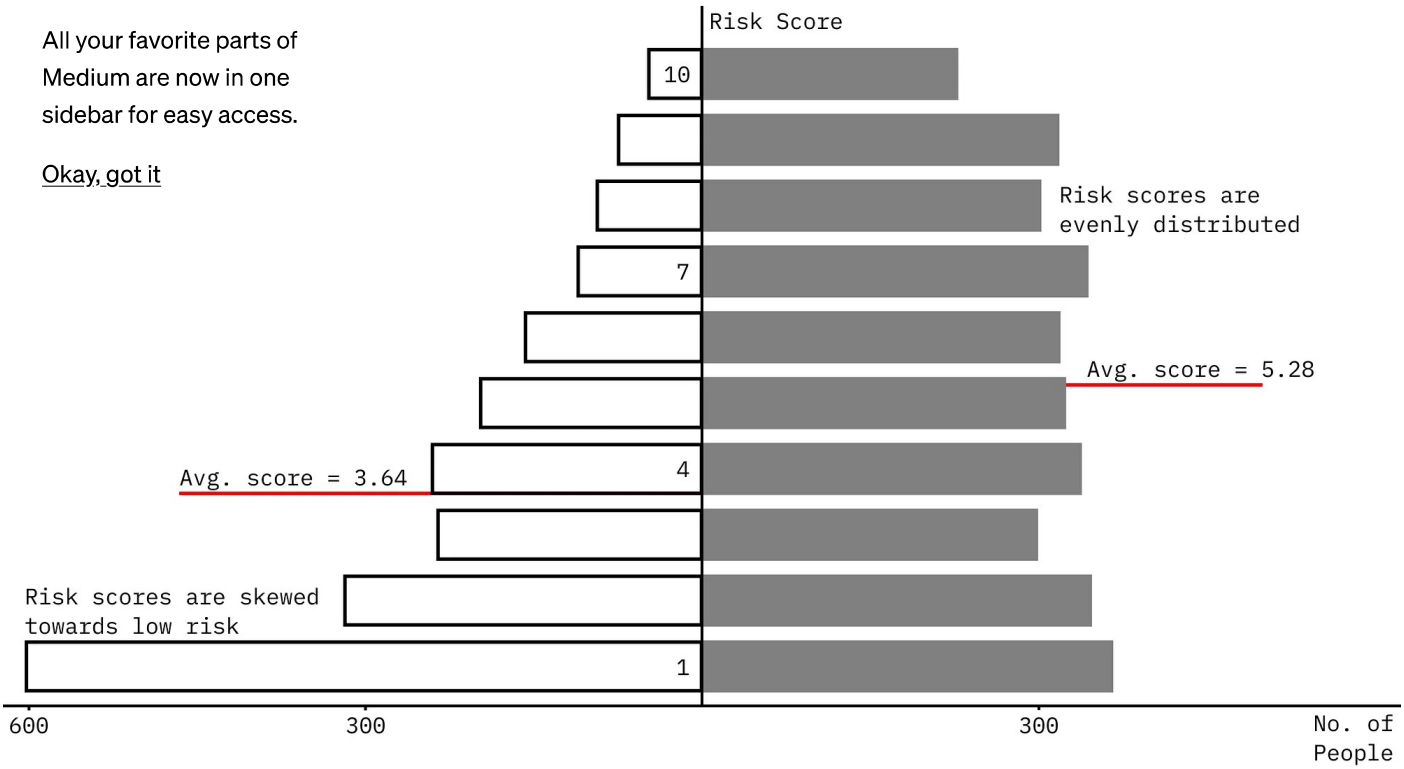
While the study analysed the algorithm's performance for general recidivism as well as violent recidivism, this article does not discusses the data and findings related to Violent Recidivism. Although the exact statistics for the two scenarios do vary, the outcomes are similar.

Higher risk scores for Black defendants

Risk scores are assigned by the algorithm according to its prediction of how likely it is for someone to re-offend. Risk scores from 1–4 are categorised into low risk, 5–7 into medium risk, and 8–10 into high risk.[1]

The average risk scores for Blacks were significantly higher than those for Whites. That was because a greater percentage of White defendants were assigned low risk scores.

> *The algorithm predicted that more Black defendants are more likely to reoffend.*



Black defendants have a higher average Risk score, with a greater number of High Risk predictions

Difference in False positive & False negative rates

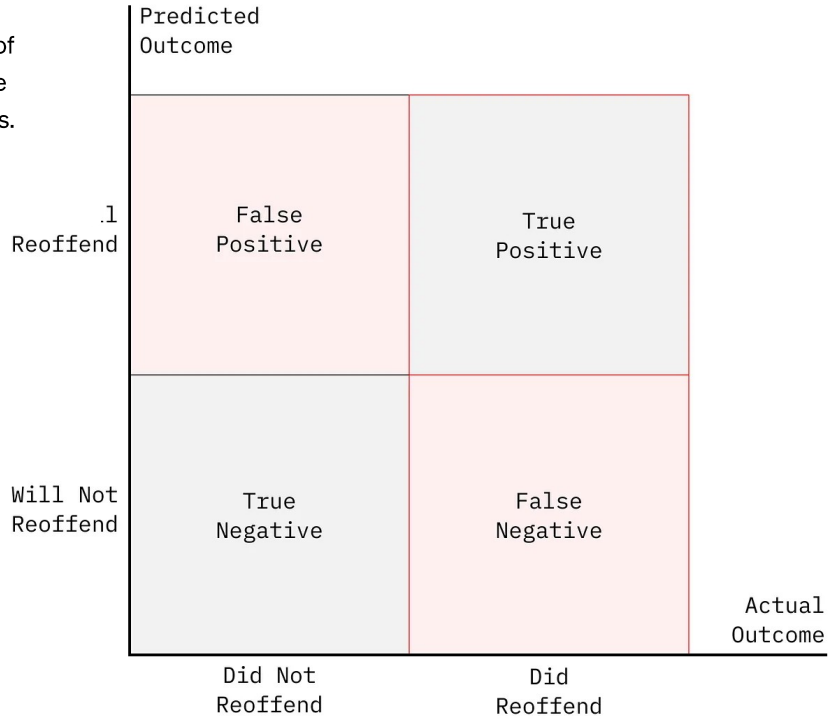
As ProPublica takes a binary approach to evaluate the algorithm's performance, (whether a prediction was true of false), we get four possible scenarios:

1. **True positives :** Predicted to re-offend and they actually do
2. **True negatives :** Predicted to not re-offend and they actually don't
3. **False positives :** Predicted to re-offend but they actually don't
4. **False negatives :** Predicted to re-offend but they actually do.

They can be graphically represented in what's called a "Confusion Matrix".

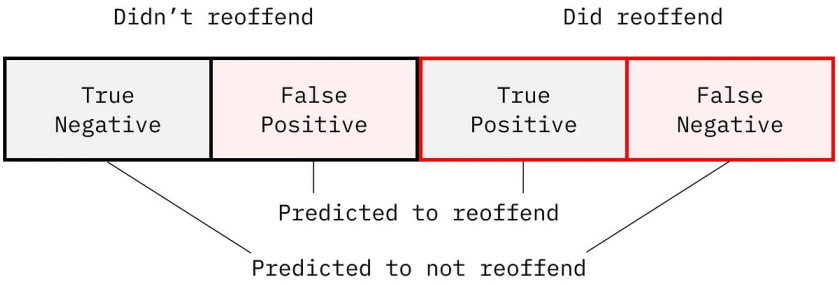
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False predictions shaded in red. Red borders for actual recidivism outcomes

For this article, let's rearrange the blocks to make them easier to discuss :



False predictions shaded in red, red borders for cases of recidivism

True positives & True negatives are cases of the algorithm's predictions being correct. ProPublica found that COMPAS was correct only ~61% of the time.[1] This accuracy rate was similar for both groups.

False positives predict an incorrectly high probability of recidivism. False negatives predict an incorrectly low probability of recidivism. When COMPAS was wrong, it was wrong in different ways for different groups.

Black defendants were more often predicted to reoffend but they didn't. White

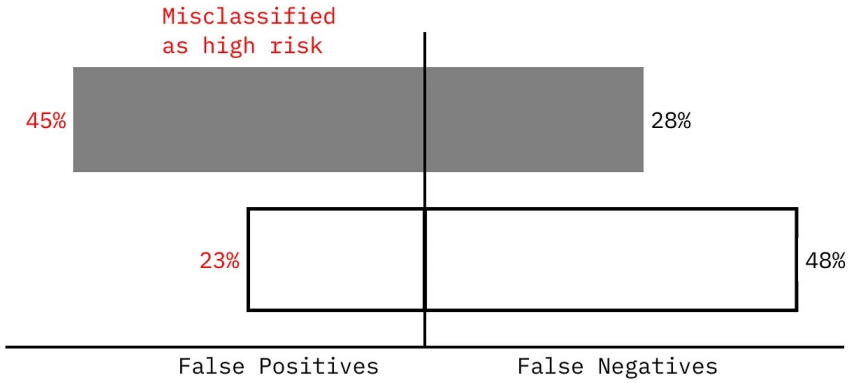
defendants were more often predicted to not reoffend but they did.

r of Whites.

rable towards Blacks.

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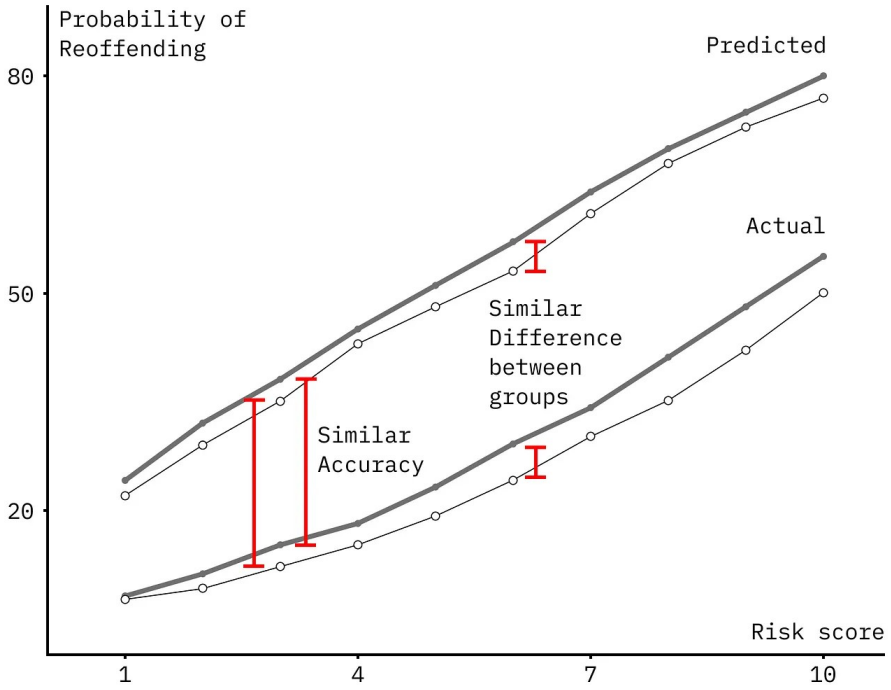
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Black defendants were twice as likely to be misclassified as high risk. White defendants were more likely to be misclassified as low risk.

But Risk scores predict probability of Recidivism equally accurately

Although the accuracy of the predictions was low, at 61%, COMPAS was similarly accurate for both groups. Moreover, the predicted probability of recidivism for any specific risk score was similar for both groups. The differences in predicted probability between the two groups reflected actual differences.



Accuracy of the predicted probability is similar for both populations. The differences in predicted probability for both groups reflects actual probabilities

That raises the question of how the results were biased in some ways but unbiased in

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lica's interpretations

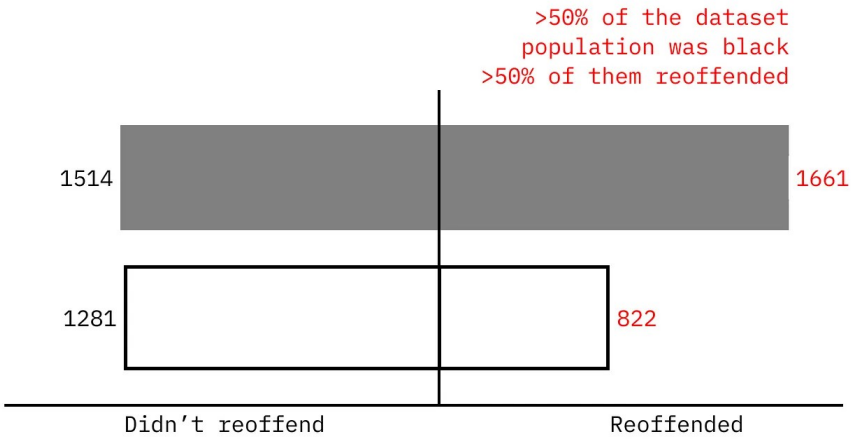
ivant), the company that owns COMPAS, raised multiple concerns drawn from the the ProPublica study. Apart from questioning the correctness and quality of the analysis, they argued that COMPAS is equally accurate at predicting risk of future recidivism for Black & White defendants, and therefore not biased. [4]

They emphasise the same point that the ProPublica authors also stated :

1. The probability of predicting future recidivism was similarly accurate for both groups.
2. Risk scores for both groups were similarly correlated to their actual probability of recidivism. In other words, Risk scores were similarly indicative of recidivism, irrespective of race.

Flores et al. (2016)[3] justify that a difference in "base rates" led to different False Positive & False Negative values for the two groups :

1. The probability of actual recidivism was higher (52% vs 39%) for Black defendants.
2. The total number of recidivists for every risk category (Low-Med-High) was higher for Black arrestees.



Greater number of people in the dataset were Black. They also re-offended at a higher rate

Who is correct here ?

Multiple articles and academic papers acknowledge that both arguments are correct, and do not contradict each other.[11][12][13] ProPublica & NorthPointe are emphasising s & metrics for evaluating fairness.

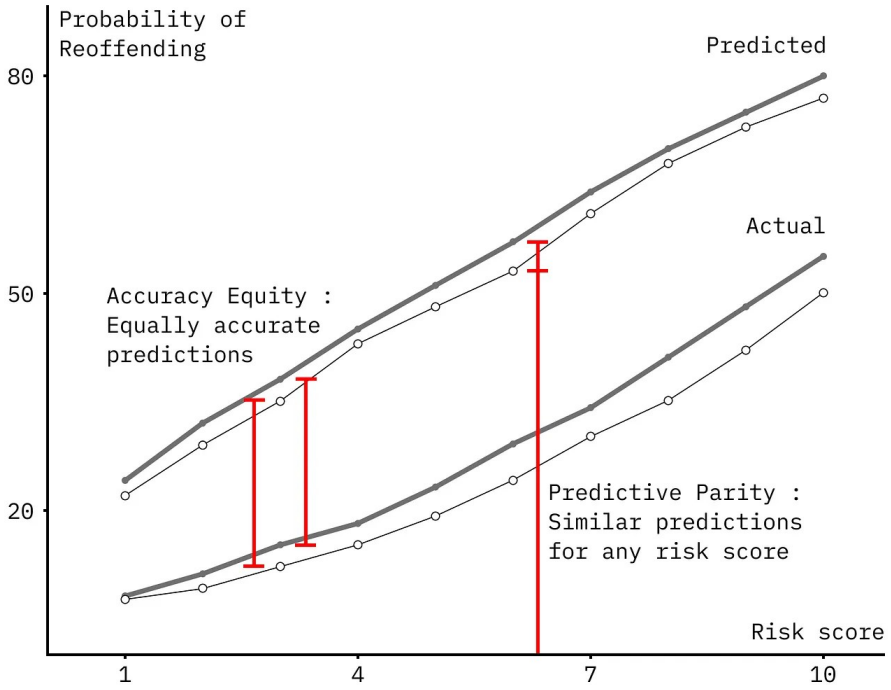
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neet one fairness criteria, but that leads to it violating another why its predictions of recidivism are equally accurate for both groups but at the same time the false rates are skewed.

By achieving similar positive (will reoffend) predictions for both groups for any specific risk score, it achieves “predictive parity”. By achieving similar accuracy for correctly predicting future recidivism, COMPAS achieves “accuracy equity”.

On the other hand, ProPublica emphasised that it fails to achieve “equalised odds” by having dissimilar false positive and false negative rates.



Equally accurate predictions for both groups. Predicted probabilities for a given risk score are similar.

Accuracy Equity, Predictive Parity & Equalised Odds

Predictive parity refers to making predictions that are independent of protected or sensitive attributes (eg. gender, ethnicity, etc.). In this context, it aims to ensure that the predicted probability of recidivism, for a specific risk score, is similar across different groups.[4]

Accuracy equity refers to the equal treatment of different groups in a dataset when considering the overall accuracy of a predictive model. In this context, it aims to ensure

that predictions about recidivism are equally accurate across different groups.[4]

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to similar error rates for different groups when making
ıtext, it aims to to make similar correct and incorrect predictions
group. Equalizing the odds means matching the true positive
rates for different groups. [5]

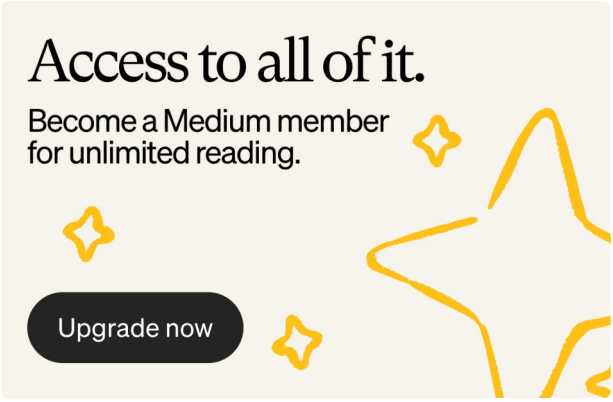
COMPAS is ensuring similar prediction accuracy for Black and White defendants, and thereby being fair via achieving “accuracy equity”. ProPublica is criticising the algorithm’s inability to achieve “equalised odds” by achieving dissimilar outcomes when the algorithm is wrong (Different false positive rates for Black & White defendants). (explored in greater detail here)

NorthPointe argue that COMPAS demonstrates accuracy equity by achieving a similar “Positive Predictive Value” (PPV) for both groups of defendants. They argue that the variation in false positive rates (FPR) is a result of differences in “base rates” of recidivism for different groups. [4]

PPV, FPR & FNR

The positive predictive value (PPV or PV+) is the probability that a person with a positive prediction will re-offend.

A high PPV means that a positive prediction (”High risk”) will result in a positive outcome (rearrest) Achieving similar PPVs for different groups serves as a measure of fairness. Let’s see how that can be explained using True Positives (TP) and False Positives (FP).



A person with a positive prediction (likely to re-offend) can fit into one of the two possibilities : The prediction turns out to be true (True positive, TP) or false (False positive, FP). PPV measures probability of true positive predictions. It can be mathematically represented as follows :

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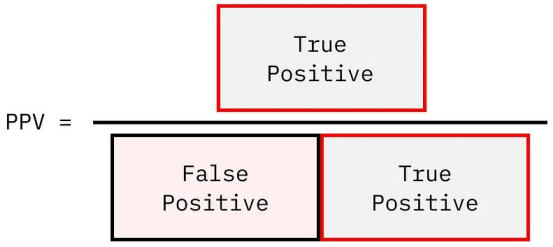
$$PPV = \frac{TP}{TP+FP}$$

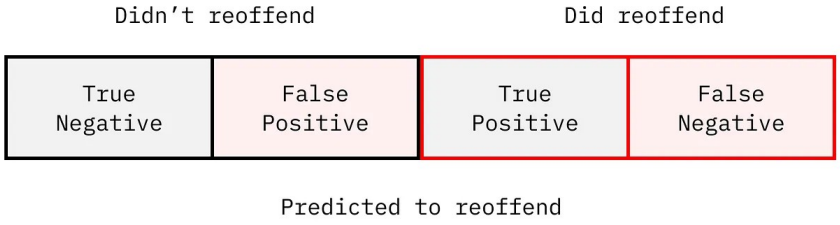
COMPAS is biased, by exhibiting different False positive (FPR) & NR) for White & Black offenders.

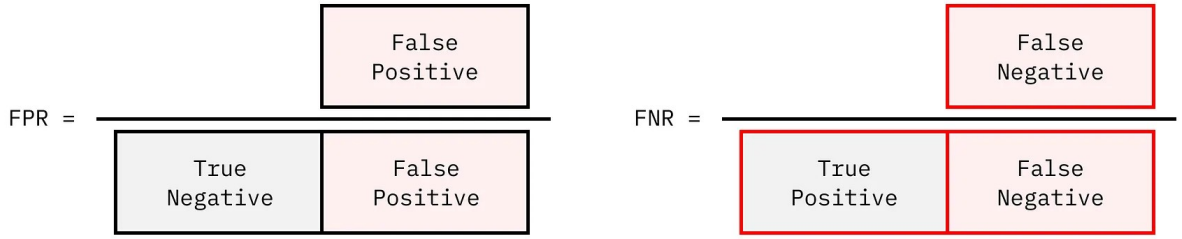
$$FPR = \frac{FP}{FP+TN}$$

$$FNR = \frac{FN}{FN+TP}$$

Visually, these can be represented in this way :







The three fairness criteria deal with different aspects of the confusion matrix

These are different fairness criteria and might not get satisfied together [9]:

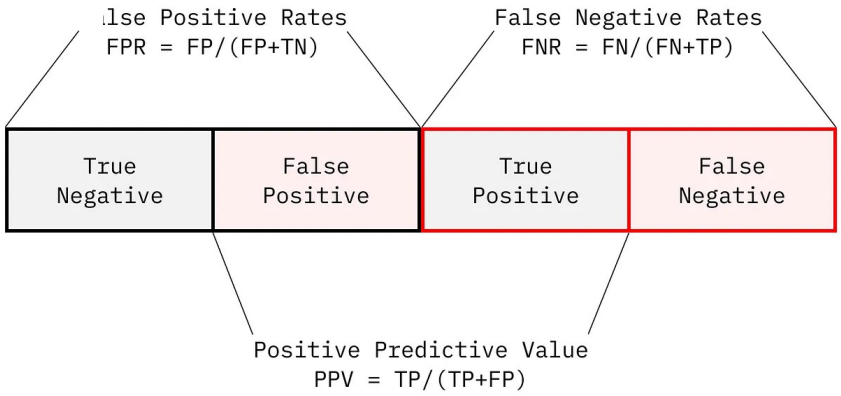
False Positive Rates only involves defendants who didn't reoffend.

ıly involves defendants who did reoffend.

ıe only involves positive predictions.

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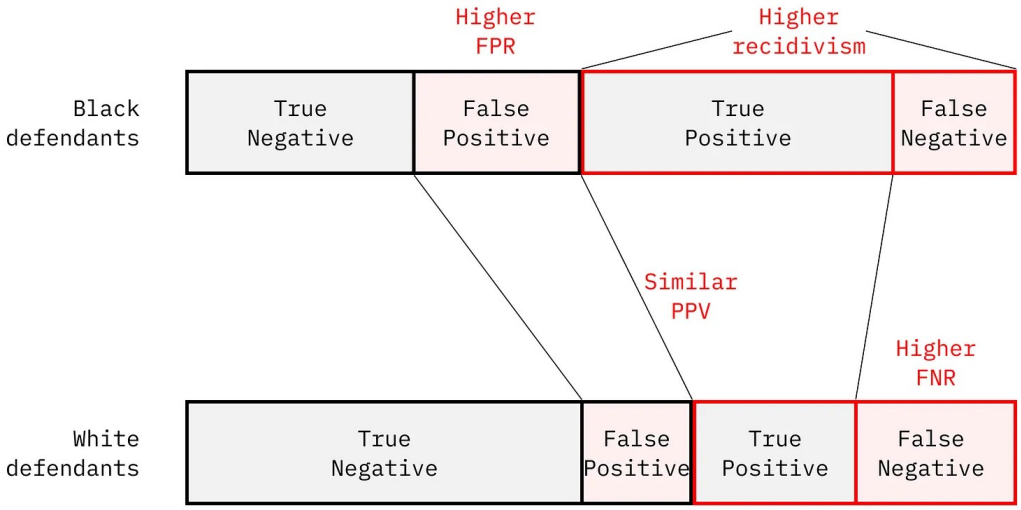
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It is possible to satisfy one on these metrics while violating the others

How different recidivism base rates affect these outcomes

A variation in base rates of recidivism in the data leads to different results.



A difference in the rate of actual recidivism leads to changes in the distribution of false and true predictions

When the algorithm tries to maintain a similar PPV, differences in base rates leads to different False Positive and False Negative rates. If the algorithm instead tried to maintain similar False Positive & Negative rates, that would lead to different PPVs for different groups.

PPV is an evaluation of whether a higher Risk score meant a greater probability of recidivism. Thus, the accuracy & validity of the Risk scores across different populations

depends on a consistent PPV.

> *…es are equally accurate for different populations, the algorithm needs to …That leads to changes in FPR & FNR.*

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…at consistent and unbiased Risk scores matter more to criminal …hen assessing an offender's probability of re-offending. Maximising the accuracy and fairness of the Risk scores would be more relevant to the context where COMPAS is used. Hence, they focused on PPV and argued that consistent Risk scores indicate a unbiased results.

They argue that it is "unrealistic" to expect similar false rates in two samples with different base rates of recidivism and risk score distributions. [4]

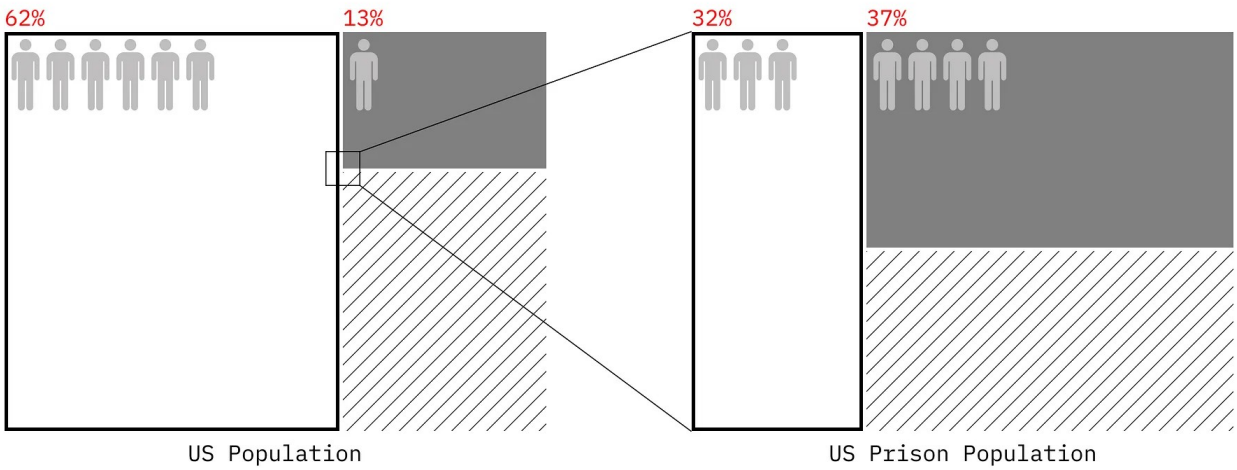
> *The dilemma :*
> *Focus on similar correct positive predictions ?*
> *Or similar errors in predictions ?*

Exploring these differences in base rates

Base rates refer to the existing probabilities of recidivism in different populations. Imbalanced base rates point to existing unequal recidivism rates for Black & White defendants. Let's explore this in some more detail.

At a national level

The population projections from the 2014 United States Census indicate that, nationally, the racial distribution among the 318 million residents was approximately 62.1% white, 13.2% black or African American, and 17.4% Hispanic. [6] In contrast, the prison population showed different proportions, with 32% categorized as white, 37% as black, and 22% as Hispanic. [7]



National statistics from 2014. Blacks have 5.5 times greater % of their total population in prison than Whites.

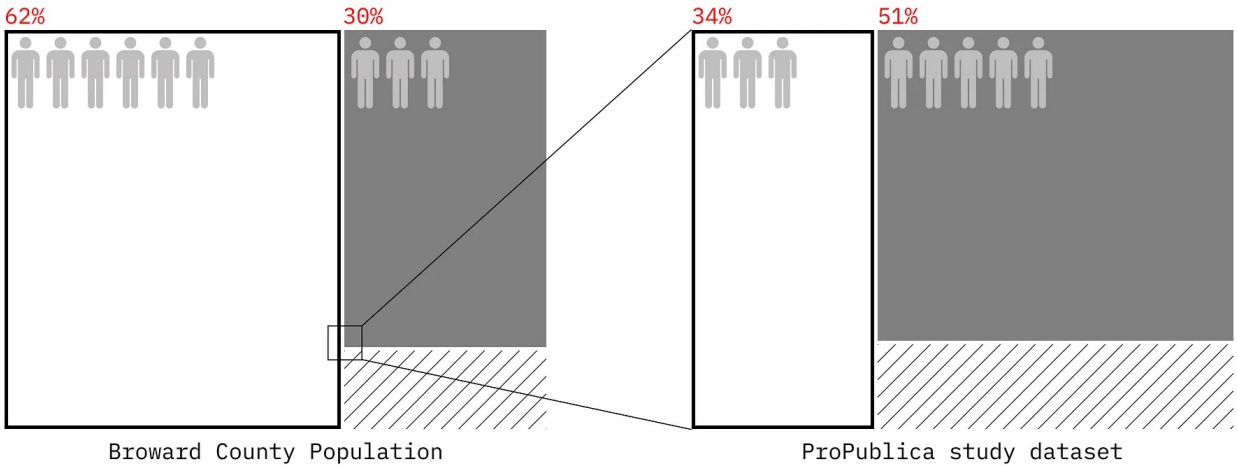
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'ents of all ages at yearend 2014, 2.7 percent of black males (or 2,724 residents) and 1.1 percent of Hispanic males (1,090 per 100,000 rving sentences of at least 1 year in prison, compared to less than 0.5 465 per 100,000 white male residents)." [7]

In Broward County, Florida

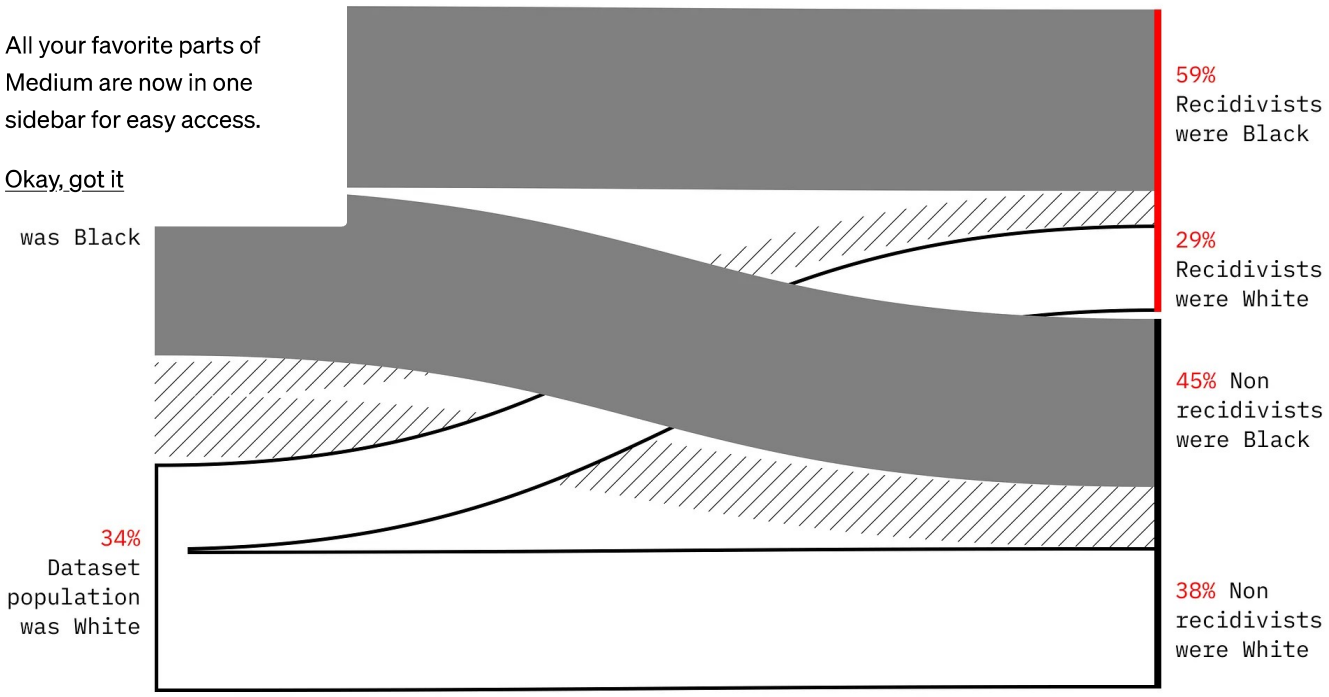
The local statistics from 2014 where ProPublica collected data for their analysis are similar :



Statistics from 2014 [8] & ProPublica's dataset

In ProPublica's dataset

If we look into the two year recidivism outcomes of the people within ProPublica's dataset, we see that apart from a larger proportion of them being Black, a larger proportion who go on to re-offend were also Black.



~45% of the individuals in the dataset re-offended (red). 59% of them were Black

Where is the difference in actual recidivism rates coming from ?

> *Black Americans are more likely to be pulled over, arrested and imprisoned.*

Instead of referring to multiple reports, articles, etc. that support this claim, Radley Balko's Washington Post article alone speaks a lot. The article summarises and links to multiple studies that demonstrate evidence of racism across various aspects of the US criminal justice system.

While this factor is not the only reason for the disparity, it highlights one aspect of the challenge of systemic disadvantages and biased treatment that populations of minority ethnicities experience.

Consequences of the algorithm's performance

From the perspective of NorthPointe & it's customers : mainly the judges who decide the fate of the arrestees, the test serves well : ~60% accuracy is "better than a coin flip" [2], and the accuracy rates are quite similar for both populations.

But from the perspective of the arrestees, particularly those wrongly classified as future criminals, the consequences are different for different groups. The more likely error for black defendants was to be wrongly classified as higher risk. That makes them incorrectly susceptible to harsher sentences, stricter treatment and overall worse outcomes.

On the other hand, white defendants were more often incorrectly classified as lower more likely to get milder corrective treatment.

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cement of existing biases.

of bias in input

The unequal proportions in the existing data make the algorithm predict a similar future outcome. In this way, although it's similarly accurate for different populations, it maintains the status quo.

> *'Predictive parity' actually corresponds to 'optimal discrimination' [12]*

Because the existing data contains greater recidivism rates for Black populations, the algorithm predicts similar future outcomes. If we're living in a reality that is biased, an algorithm that is designed to be able to accurately predict this reality, will end up making biased predictions.

Bias in the future as a result of bias in the past ?

Researchers claim that algorithmic systems can help us address the current biases in the criminal justice system :

> *Risk assessment tools informed by objective data can help reduce racial bias from its current level. It would be a shame if policymakers mistakenly thought that risk assessment tools were somehow worse than the status quo.[3]*
>
> *If used properly, criminal-justice algorithms offer "the chance of a generation, and perhaps a lifetime, to reform sentencing and unwind mass incarceration in a scientific way."[10]*

But while there is potential in these approaches, the significantly low prediction inaccuracies & difficult to address biases they exhibit, makes it seem that this potential is yet to be realised.

References :

1. https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm
2. https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing
3. https://www.uscourts.gov/federal-probation-journal/2016/09/false-positives-false-negatives-and-false-analyses-rejoinder

4. https://www.equivant.com/response-to-propublica-demonstrating-accuracy-equity-and-predictive-parity/

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ı/courses/res-ec-001-exploring-fairness-in-machine-learning-for-opment-spring-2020/pages/module-three-framework/fairness-

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8. https://datacommons.org/place/geoId/12011?category=Demographics#Population-by-race
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10. https://www.americanbar.org/groups/public_education/resources/law_related_education_network/how_courts_work/casediagram/
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12. https://www.propublica.org/article/bias-in-criminal-risk-scores-is-mathematically-inevitable-researchers-say

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Kate Harwood
Nov 21, 2024 (edited)

As someone who teaches AI Ethics, I have been looking for a succinct, thorough summary of this debate that delves into both sides of the analysis for years, and have been unable to find anything until I stumbled upon this. Thank you!

1 Reply

Kiron Ravindran
Oct 14

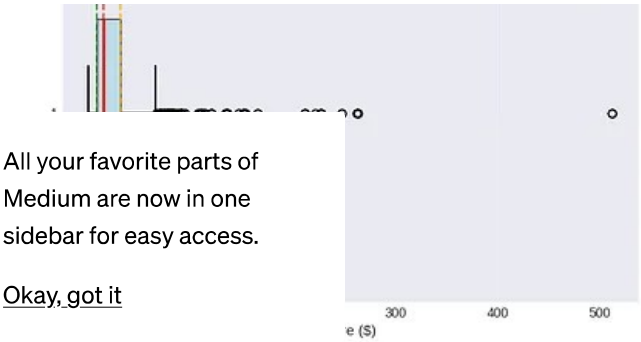
Thank you for this brilliant explanation and for keeping it in the public domain

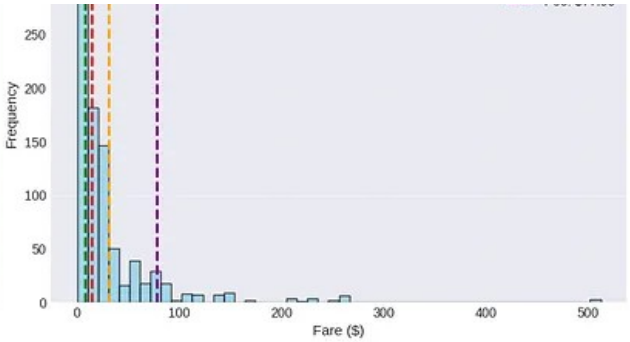
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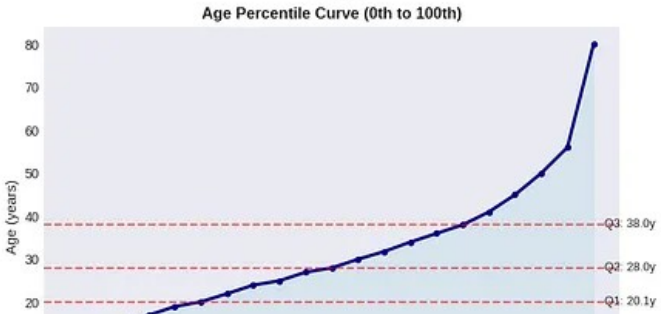
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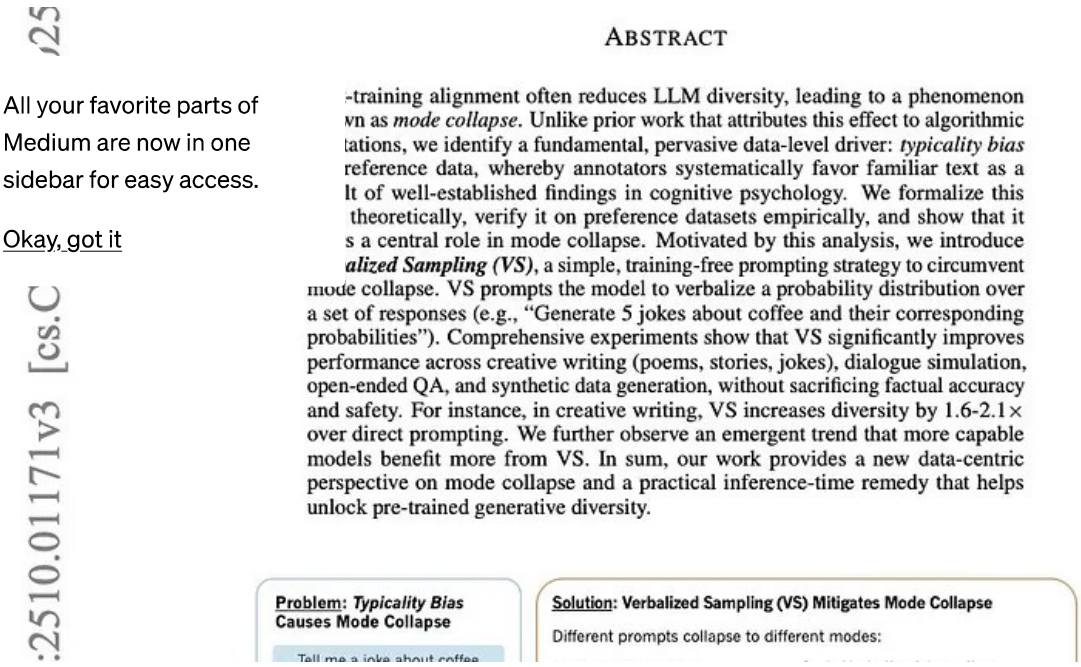
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ABSTRACT

-training alignment often reduces LLM diversity, leading to a phenomenon vn as *mode collapse*. Unlike prior work that attributes this effect to algorithmic tations, we identify a fundamental, pervasive data-level driver: *typicality bias* reference data, whereby annotators systematically favor familiar text as a lt of well-established findings in cognitive psychology. We formalize this theoretically, verify it on preference datasets empirically, and show that it s a central role in mode collapse. Motivated by this analysis, we introduce ***alized Sampling (VS)***, a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., "Generate 5 jokes about coffee and their corresponding probabilities"). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by 1.6-2.1× over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

Problem: *Typicality Bias* Causes Mode Collapse

Solution: Verbalized Sampling (VS) Mitigates Mode Collapse

Different prompts collapse to different modes:

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